DoS Incident Response Analysis Using NIST CSF

Course: Google Cybersecurity Professional Certificate
Project Type: Incident Response Analysis
Scenario: Simulated Denial-of-Service (DoS) Attack

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1. Project Overview

This project involved analyzing a simulated denial-of-service (DoS) attack against a multimedia company's internal network.

The objective was to identify the cause and impact of the incident and develop a security response plan using the National Institute of Standards and Technology Cybersecurity Framework (NIST CSF).

2. Incident Report Analysis

Summary

A malicious actor launched a denial-of-service (DoS) attack by flooding the company's network with ICMP packets through an improperly configured firewall.

The excessive traffic overwhelmed the network, preventing normal internal traffic from accessing network resources. The disruption lasted approximately two hours.

Identify

Attack Type: Denial-of-Service (DoS) attack using ICMP packet flooding.

Affected Assets: The company's internal network and network-connected resources.

Identified Vulnerability: An improperly configured firewall allowed excessive incoming ICMP traffic to overwhelm the network.

Protect

The following security measures were identified to strengthen network protection:

- Implement source IP address verification on the firewall to identify potentially spoofed IP addresses.
- Deploy an intrusion detection and prevention system (IDS/IPS) to filter suspicious traffic.

Detect

The following measures were identified to improve the detection of future attacks:

- Configure firewall rules to limit the rate of incoming ICMP packets.
- Implement network monitoring software to detect abnormal traffic patterns.

Respond

The following response procedures were proposed for similar incidents:

1. Block malicious incoming ICMP packets.
2. Stop non-critical network services to limit the impact.
3. Keep or restore critical services needed for business operations.
4. Analyze firewall, IDS/IPS, and network logs.
5. Determine the source and scope of the attack.
6. Document the incident and response actions.

Recover

The following recovery procedures were proposed:

1. Restore affected network services.
2. Verify that network resources are functioning normally.
3. Confirm that firewall and IDS/IPS configurations are properly configured.
4. Review the cause of the incident.
5. Update security procedures and configurations to reduce the likelihood of similar attacks.

3. Key Findings

The incident demonstrated how an improperly configured firewall can expose an organization to network-based denial-of-service attacks.

The analysis highlighted the importance of traffic filtering, continuous network monitoring, incident containment, and recovery procedures.

4. Skills Demonstrated

- Network Security
- Incident Response Planning

- NIST Cybersecurity Framework
- Denial-of-Service Attack Analysis
- Firewall Configuration Concepts
- IDS/IPS
- Network Traffic Monitoring
- Security Incident Documentation

5. Conclusion

This project provided practical experience analyzing a simulated network security incident and developing a structured incident response plan.

Applying the NIST Cybersecurity Framework helped organize security measures for identifying risks, protecting network resources, detecting suspicious activity, responding to incidents, and restoring normal operations.

This project was completed as part of the Google Cybersecurity Professional Certificate using a simulated security incident. The incident analysis and proposed response procedures are based on my completed coursework.